A Review of Radiation, GR, and Pulsars, Oh My!

Ashwati Sanjay
Professor: Andrzej Prus-Czarnecki

1 Introduction

The discovery of the first binary pulsar by Russell A. Hulse and Joseph H. Taylor was integral to developing the field of gravity theories[1]. It provided the first strong-field testing grounds for the predictions of Einstein's general relativity (GR) and alternatives, prior to which only weak-field testing was accessible within the solar system. The development of ground-based observatories followed and radiation due to gravitational waves (GW) was detected in 2015[2].

The Hulse-Taylor (HT) pulsar is a binary consisting of two neutron stars (NS) where the dominant effect between the two is their gravity. As the two objects orbit around each other, they lose energy and the distance between them closes and this also shows as the periastron time decreasing. Both of these effects can be explained as a consequence of energy being lost from the system in the form of radiation of GWs, a phenomenon predicted by Einstein's GR[3]. This was crucial evidence of GWs and key in bolstering the reputability of GR.

Many other effects can be observed that are intimately tied to various properties of GR such as the precession of the periastron, certain interactions between the primary and companion in a binary, and the signals that are detected.

2 Theory

During her PhD studies, Jocelyn Bell Burnell was on the hunt for quasars which are the bright cores of massive galaxies that are powered by supermassive black holes. While scouring through the endless paper charts, she noticed a strange "scruff" in the data. It was a weird signal which was initially dismissed, until it showed up again in the charts at consistent intervals. In 1967, the discovery of the radio pulsar, PSR B1919+21, came as a surprise[4]. It was not being looked for and yet it made its presence known with a distinct signal, a burst of high intensity radio waves. This finding paved the way for gravity theories to become more than just equations on paper, but something that can be tested in a tangible manner. But what is a pulsar, and why are they interesting in the first place?

A pulsar is a late-stage stellar object, the end result of the dazzling and brief life of a massive star, one that went supernova and turned into a neutron star (NS). This NS would produce a beam of radiation, that we detect as "pulses" of radio emissions. The beam sweeps across the sky as the NS rotates about its axis, and in the case that the beam points towards Earth, we get the chance to study it. The immediately surprising aspect about pulsars is the stability in their period, with the pulsar PSR B1919+21 having a spin period of 1.3372795 ± 0.0000020 s[4]. Because of their stability, they can function as accurate clocks in the universe[5].

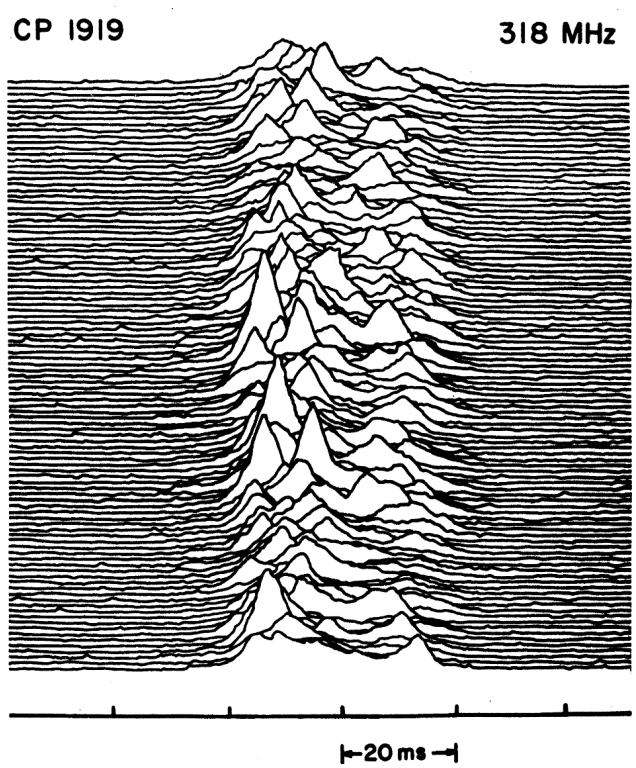

Figure 1: Shows the pulse profiles for the pulsar PSR B1919+21 in radio. The horizontal axis is the spin phase in high time resolution while the vertical axis represents the intensity of the radio emission. Many pulse profiles are stacked on top of each other to get the average profile, here they are displaced for viewing purposes[6][5].

There are many notable aspects, for one being the Doppler shift the signal experiences due to the object being so far away. As it travels through debris and around various obstacles, the signal we receive is Doppler shifted, where the lower frequency radiation has a delay before reaching our detectors[5][7]. The appearance of these signals is that of a generally flat line except when the magnetic pole points towards the Earth, in which case there's a distinct bump or "pulse" in the reading. This is called the pulse profile of the pulsar as in Fig. 1, where there is a window of high intensity emission, the pulse in question, that comes regularly. The shape of the pulse varies, with a mountain range appearance, but is clearly higher than background and this part is generally constant from collection to collection.

Initially these signals were attributed to white dwarfs (WD) but the discovery of the Vela and Crab pulsars with their incredibly short periodicity (milliseconds) that slowed down over time as well as being inside supernova (SN) remnants, which points to these being end-stage products of stellar evolution, fit the models that described NSs and led to their acceptance. They are formed when a massive star undergoes SN and leaves it's immensely dense core behind, this being what we refer to as a NS. Since their densities are supranuclear, there is a great lack in the understanding of the equation of state (EoS) that describes them. The uncertainty is on the order of O(20%) for things such as maximum mass, radius, and moment of inertia (MoI), which means that this is a great area of research[8].

The radio emission from pulsars still remains poorly understood but are used in testing nonetheless. They emit broadband radiation but due to the higher frequencies being much weaker in power output, in the sky they appear mainly as point sources of radio emissions without any spectral features. The lighthouse model describes a rotating magnetized NS, where the spin axis is misaligned with the axis of magnetization. The majority of the radio emissions come from open magnetic field lines near the poles and as the NS rotates, the beam sweep across the sky. If the beam faces Earth during its rotation, we can potentially see these emissions and they appear as a "pulse" where the intensity increases periodically, similar to a lighthouse. The majority of pulsars are studied using time-domain data, where a radio telescope is pointed towards the small section of sky where the pulsar is and records the intensity of light received from that section over some course of time. This is unlike using other methods which typically require high image resolution, where all that is required in this case is high time resolution and a sensitive radio telescope, which is relatively easier to achieve.

Using the lighthouse model, the number of rotations of the pulsar can be measured and there would be a change in the spin over time. Looking at Fig. 1, it is obvious that each individual pulse varies from detection to detection. This can also make it difficult to get the timing correct. By stacking many pulse profiles on top of each other, the signal to noise ratio (SNR) increases and helps make timing more accurate. This process is called folding and is vital to make sense of the signal[9]l. In order to fold a time series, there needs to be an accurate estimate of the spin frequency of the pulsar in the frame of the radio telescope. This can be predicted using a timing program given a reasonable timing model and a description of the motion of the radio telescope relative to the solar system barycentre (SSB). This is particularly accurate when describing recycled pulsars[10].

$$N(T) = \frac{\phi(T) - \phi_0}{2\pi} = \nu(T - T_0) + 0.5\dot{\nu}(T - T_0)^2 \tag{1}$$

From a high SNR pulse profile, a longitude on the pulsar can be chosen as the zero phase position and is kept consistent. Then lining all the pulse profiles up with their zero phase and folding, the time of arrival (ToA) of the pulse can be calculated. The ToA can be finicky as there can be dispersion in the beam which leads to frequency noise, or if the SNR is too high it can be sensitive to the randomness in the individual pulse profile but if the profile is too smooth, the timing will be poor[5].

From a good pulse profile, there are two types of pulsars that can be detected as in Fig. 2. The normal pulsar is alone and eventually their spin slows and the emissions will become less detectable, becoming invisible to us. In the case of a binary, there are many forms that they may take.

In a binary star system the more massive of the two, the primary, will eventually become a giant and go SN, disturbing the environment around it and potentially kicking the companion out, leading them to both be unbound. It will then become a NS, producing emissions as

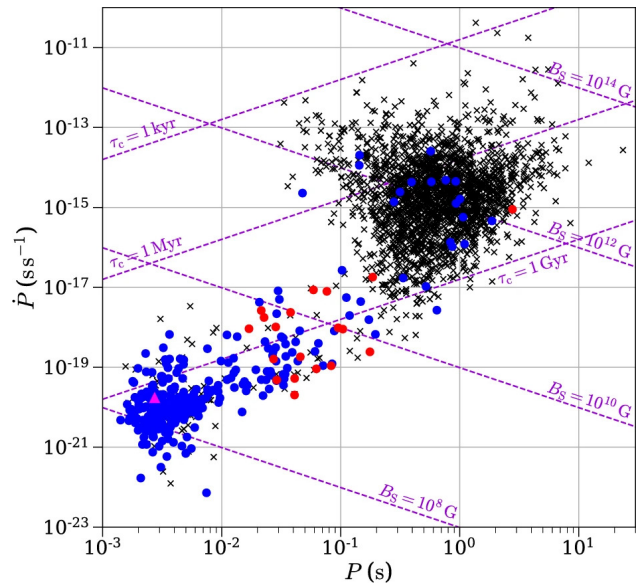

Figure 2: The black crosses are isolated, "normal", pulsars. These have higher spin periods that change faster, as well as stronger magnetic fields and are younger. The blue circles are binaries and the red circles and double neutron star (DNS) binaries. They get spun up, so most of this group includes the recycled, quicker spinning pulsars. They are generally older and had their magnetic fields stripped away. The magenta triangle is a triple system. B_s is the surface magnetic field and τ_c is the characteristic age[11].

a normal pulsar would and eventually slowing down. If the companion is still bound, it will evolve and reach its giant phase, filling its Roche lobes and matter can be transferred from the companion to the primary. During the common envelope phase, the pulsar gets spun up from the accretion and the spin aligns with the orbital angular momentum and it is thought that during this, the magnetic flux density of the NS gets stripped. The recycled pulsar describes this situation, when the companion accretes matter onto the primary, spinning it up such that the spin is much higher than a regular pulsar, often becoming millisecond pulsars (MSPs)[10]. If the companion is light it turns into a WD with the orbital eccentricity staying low. If heavy, the companion goes SN, further disrupting the environment, potentially becoming unbound, and becomes a NS. Due to the many disturbances faced by a binary pulsar system, there are few to study. In a double neutron star (DNS) system, because of the energy loss the two objects get closer on a shorter time scale than regular pulsar-binaries and GWs can be observed.

3 Method

Many tests involving pulsars utilise the incredible consistency of the periodicity and the accuracy of the timing data collected. This is especially true for recycled pulsars, where the fast spin contribute to the preciseness. Characteristics of the pulsar can be determined from the ToA, which can be estimated from the pulsar ephemeris, which is the position over time

data, and equation 1 which can be determined using a timing program. The optimal timing parameters are found using the least-squares method on the normalized residuals:

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{T_i - T_{i,p}}{\sigma_i} \right)^2 \tag{2}$$

From equation 1, a small error in either ν or $\dot{\nu}$ will produce a linearly or quadratically increasing residual respectively, and so those values can be adjusted for until there are no such effects on the residual.

When the HT pulsar was discovered, there were no models to describe them, so the creation of these timing programs was integral to the development of this field. Several timing models have since been developed across the last few decades, where the essential difference is the addition of relativistic corrections and the choice of coordinates. For a binary system, the following Doppler shifted equation can be used:

$$T = \tau_{obs} - \Delta_{E,\odot} - t_0 - \Delta_{R,\odot} - \Delta_{S,\odot} - \Delta t_{dis}(F) - \Delta_R - \Delta_S - \Delta_E - \Delta_A \tag{3}$$

[12]
There are three terms relating to the delays within the solar system (denoted with subscript $\odot$) and four terms relating to the delays from the binary. Using the relativistic Doppler factor D, quantities with units of mass, length, or time can be easily converted from the reference frame of the binary into the frame of the SSB by a factor $\frac{1}{D}$.

The $\Delta_{E,\odot}$ and Δ_E relate to the Einstein delay for the solar system and binary respectively. This means that regarding the Solar System, because the observer is within the the gravitational field of the Sun, it experiences time dilation as the Earth travels around the SSB. It also experiences gravitational redshift due to the other masses within the system. The solar system term comes from the four-dimensional transformation between the coordinate system of the SSB and the observer[12]. For the binary, it works similarly where again time dilation and gravitational redshift is experienced due to the motion of the pulsar in the centre-of-mass (CoM) frame of the binary. This effect is independent of how the binary is oriented and only occurs when ω changes drastically[13].

The $\Delta_{R,\odot}$ and Δ_R terms come from the Rømer delay. This delay relates to the changing distance between the pulsar and observer due to the relative motion, and can be adjusted for by finding the projection of the observer on the incoming emission in the frame of the SSB. Two vectors are enough to describe this, one being the position of the observer relative to the centre of the Earth, and the other being the vector from the centre of the Earth to the SSB, where the orientation of Earth is also taken into account[12]. In the binary, again the effect is similar, where the relative motion is also effected by the pulsar orbiting the binary's CoM[5].

The $\Delta_{S,\odot}$ and Δ_S terms account for the Shapiro delay. This is caused by light propagating through curved spacetime, where in the Solar System it would be affected by the mass of the Sun and planets, and in the binary it would be effected by the mass of the binary and

whether the companion is in the line of sight, so if a system is edge-on the Shapiro delay will be much larger. This can be calculated once α and δ are specified[12][14].

The relativistic aberration delay, Δ_A, is specific to the binary. It is due to the spin of the pulsar and how that rotation affects the emission beam, leading to the ToA to be shifted and the shape of the profile to change. The orbital motion around the CoM also plays a part in the bending of the emission beam, and either more or less time is needed to compensate for this rotation[12]. It wouldn't be an issue if the pulsar was a non-rotating, pulsing source similar to stars near the end of their life. The magnitude of this effect can also be predicted by GR calculations[15].

Similar to how the ToA parameters ν and $\dot{\nu}$ can be manipulated to minimize the residuals, the same can be done when accounting for the motion of the Earth, where if it is incorrectly subtracted, a sinusoidal trend appears in the residuals and the values of α and δ can be adjusted[12]. If there is a growing sinusoid, the proper motion of the pulsar ($\mu_\alpha = \dot{\alpha} cos\delta$ and $\mu_\delta = \dot{\delta}$) can be derived by fitting $\dot{\alpha}$ and $\dot{\delta}$. If we're lucky, the pulsar will be close enough such that the distance can be found using parallax[5].

The Doppler shift technique, while helpful, has its limitations. When trying to constrain the orbital motion, the binary-related delays must be taken into account. Once an accurate ephemeris is made, the precision increases by thousands to millions, meaning any small relativistic effect will be present, which is why pulsar binaries are amazing laboratories for gravity theories[12]. Another important effect is the gravitational radiation and the subsequent orbital decay. When a binary sits close together, or has a very eccentric orbit, it has the chance to produce strong radiation through GWs, therefore losing energy and leading to an increase in the orbital period following Kepler's third law, which shortens with each pass-by. The effect is quadratic where as they get closer, the ToAs get smaller over time[16].

Following from this, due to the low precision of Newton's gravitational constant, we take it upon ourselves to restructure Kepler's third law[17]. With the rewriting of the semi-major axis to be in terms of time units, $\bar{a} \equiv a/c$:

$$\left(\frac{P_b}{2\pi}\right)^2 = \frac{\bar{a}^3}{T_\odot M} \tag{4}$$

We can write the mass to be in terms of solar mass units by defining $T_\odot \equiv \frac{(GM)_\odot^N}{c^3} = 4.925490947641...$ μs as the nominal Sun's solar mass parameter in time units[17]. Now the precision is tied to parameters that can be measured rather than G. Now the semi-major axis can be determined independently of the orbital inclination i:

$$\bar{a} = \left(\frac{P_b}{2\pi}\right)^{2/3} (T_\odot M)^{1/3} \tag{5}$$

The mass ratio R can be expressed as[17]:

$$R \equiv \frac{M_p}{M_c} = \frac{a_c}{a_p} = \frac{x_c}{x} \tag{6}$$

with $x_c = (a_c/c)sin(i)$, and $a = a_p(M/M_c)$. From this the Newtonian mass function can be derived[17]:

$$f(M_p, M_c, i) = \frac{(M_c sini)^3}{(M_p + M_c)^2} = \frac{x^3}{T_\odot}\left(\frac{P_b}{2\pi}\right) \tag{7}$$

In most cases, two of the three unknowns must be assumed to obtain a lower mass limit of the companion M_c, with $i = 90°$ and some assumed mass of the primary, M_p. R can be measured if the companion is optically bright, and a value for x_c can be calculated. To find M_c, if the companion is an optically bright WD, the surface gravity can be determined by using the width of the Balmer lines and combined with the mass-radius relation to find the mass and radius of the WD[12]. However, both are more often than not unavailable and require many assumptions and models to estimate reasonable values. If the first post-Newtonian (PN) corrections are included, the relation $a = a_p + a_c$ and equation 7 hold in any Lorentz-invariant gravity theory, such as GR. For alternative theories to GR M_c is important as it is determined in the Newtonian-limit. Equation 8 is valid for almost all binary pulsar systems tested excluding the double pulsar case[12].

The post-Keplerian (PK) parameters can hold different values depending on the theory it is being investigated within. In GR specifically, they depend on the regular Keplerian parameters (P_b, e, and for one case x)[12], and the masses of the primary and companion. All the equations describing them are listed in the Appendix, equations 11-17[13]. It is notable the simplicity that the equations for the PK parameters take, owing it to a property of GR mentioned previously where the internal structure of an object holds no relevance, and it holds for many body systems. By simply measuring two PK parameters, the masses of both objects in the binary can be calculated, plus the inclination angle i can be found either explicitly or implicitly from the masses[12]. A theory can be tested based on the agreement of the PK parameters in the mass-mass space. The leading order terms suffice for the typical pulsar-binary but in the case of the double pulsar, higher order terms are relevant when describing the timing data[18].

The orbital period and its time derivative can be affected by a few different factors. First, the galactic acceleration of the pulsar and the SSB vary since their placement in the gravitational field of the Milky Way are likely different. However, this can be estimated by modelling what the field looks like at the position of the pulsar in the MW. In some instances there may be other fields of comparable weight to keep in mind, such as local stars or globular clusters[12]. Another effect is the Shklovskii effect, where the rotation of the pulsar appears to slow down due to its proper motion. In order to correct this, d and μ need to be measured as both can limit the precision of $\dot{P}_b$. They both are present as the second and third terms respectively in the equation below describing the orbital period[12].

$$\left(\frac{\dot{P}_b}{P_b}\right)_{int} = \left(\frac{\dot{P}_b}{P_b}\right)_{obs} - \frac{\mathbf{a}_{CM} \cdot \hat{\mathbf{K}}_0}{c} - \frac{\mu^2 d}{c} \tag{8}$$

Curved spacetime also leads to an effect on the spin of the pulsar, where a free-falling object has some intrinsic spin, but it appears to be different to an outside observer. This is mainly due to the geodetic precession, leading to spin-orbit coupling and so the pulsar precesses around the conserved total angular momentum of the system at a rate Ω_p^{geod} depending on the masses and the Keplerian parameters. The measurement of this can be used as a test for GR[18].

$$\Omega_p^{geod} = 0.5 \left(\frac{P_b}{2\pi}\right)^{-5/3} (T_\odot M)^{2/3} (1-e^2)^{-1} (3+X_p) X_c = \frac{1}{6}(3+X_p) X_c \dot{\omega} \tag{9}$$

If the spin axis precesses there is a change in the part of the emission beam seen from Earth which leads to a change in the pulse profile, where it can potentially become undetectable[18]. The PK parameters are elusive, they have many requirements in order to measure even one and the biggest obstacle is the faintness of the pulsar which limits the timing precision. The only way to get around this is to build more sensitive radio telescopes. To measure $\dot{\omega}$ and γ, the orbit must be quite eccentric, and γ also needs a large change in ω during the viewing period. The fractional uncertainty decreases as $\Delta T^{-3/2}$ where ΔT is the timing baseline, but for γ this decrease slows after a large fraction of the precession cycle is completed. If the conditions aren't met, these variables aren't separable from the Keplerian parameters[12][13]. The parameters for the Shapiro delay (r, s or h_3, ζ) are better evaluated when the binary is edge-on, as this is when the effect is the largest. They also depend on the timing precision since the delay is a periodic effect. The uncertainty of the parameters decreases according to $\Delta T^{-1/2}$. In order to separate the Shapiro delay from the Rømer delay, it helps if the orbit is eccentric and precesses[12][14].

To measure $\dot{P}_b$, short orbital periods are preferred with GWs contributing on the order of $P_b^{-5/3}$ and the kinematic portion scales with P_b. Detecting δ_θ requires similar circumstances as γ and also an incredibly short orbital period. The fractional uncertainty of $\dot{P}_b$ and δ_θ decreases with $\Delta T^{-5/2}$ and again as similar to γ, the uncertainty in δ_θ slows after a significant portion of the precession cycle[12][18].

Geodetic precession needs a pulsar's spin to be very misaligned with the orbital angular momentum, and this is mainly only seen in DNS systems. To measure the Lense-Thirring (LT) contribution to di/dt, a large NS spin is required whereas for the LT contribution to $\dot{\omega}$, the mass needs to be incredibly precise. For the want of short spins, MSPs are a great candidate for accurate timing precision, but they're usually in a circular, wider orbit so not all parameters can be measured. This also affects calculating the Shapiro delay, since their companions are usually of low mass, and the geodetic precession, since their spins typically align well with the orbit angular momentum. So taking into account the very demanding requirements for the PK parameters, DNS systems are of big importance for gravity tests[12][18].

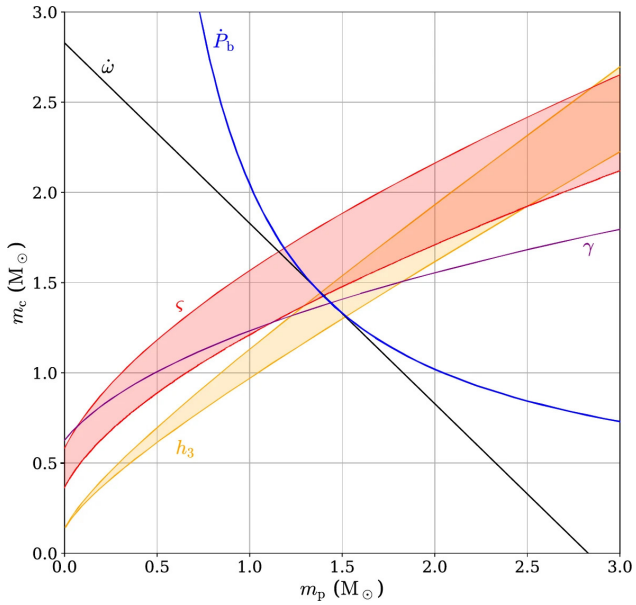

Figure 3: HT Pulsar: The three PK parameters agree on a subsection of the mass-mass diagram assuming GR, confirming existence of GWs. Shapiro delay eventually detected, but some uncertainty in distance[19].

4 Results

Binary pulsar systems, especially pulsars with compact companions, provide some of the best testing grounds for GR in a strong gravitational field, where the spacetime deviates significantly from Minkowski spacetime[12]. They can be viewed as point-like sources where the distance between them is large compared to their size. The main test is measuring as many PK parameters as possible and checking if they agree on similar mass ranges for the primary and companion and if they do, GR has passed the test[18].

Going back to the HT pulsar with a spin period of 59 ms and an eccentric orbit ($e = 0.62$) with period 7.75h and an unseen compact companion, the masses for both can be deduced assuming GR which yields comparable masses for both, almost $1.5M_\odot$, meaning it is highly likely the companion is also a NS. During the time it was observed, there was a noticeable change in the periastron, advancing by roughly 4.2° per year. Later the the decay in the orbital period $\dot{P}_b$ was also measured[1][3]. The three PK parameters agreed on a subspace of the mass-mass plot by assuming GR (Fig. 3), and so confirmed the existence of GWs which is a prediction by GR, and showed that it is a valid theory to describe two massive objects in a strong field. The distance to the binary was uncertain however, so corrections involving extrinsic contributions to $\dot{P}_b$ is limited. It was also found to undergo geodetic precession, causing changes in the emission geometry and again, played an important role in the development of GW detectors on the ground[20][12].

The double pulsar system discovered in 2003, PSR J0737-3039A/B, is also the only double pulsar system found thus far. Both pulsars were detectable at one point, with the primary

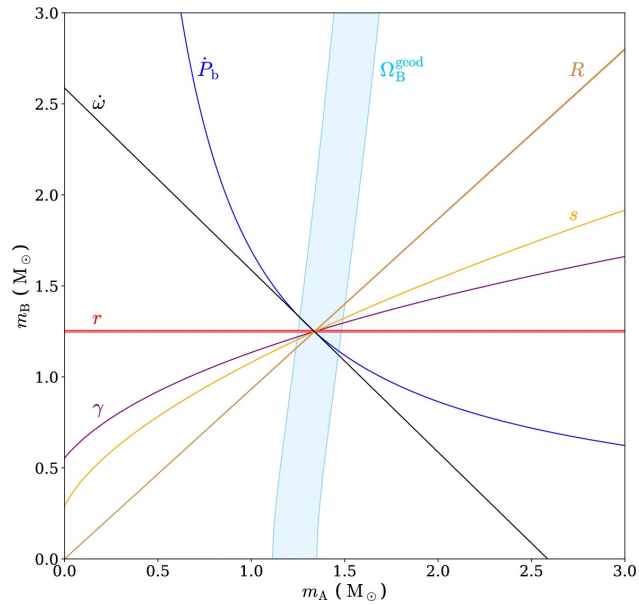

Figure 4: Mass-mass diagram of the double pulsar, with five PK parameters along with mass ratio R and geodetic precession Ω_c^{geod}. All curves agree on a region assuming GR, and is therefore self-consistent. Figure from [18], values of Ω_c^{geod} from [18].

being a recycled MSP and the companion being non-recycled. The system is seen almost edge-on so a strong Shapiro delay is present, and the masses are smaller than those in the HT pulsar but it is more relativistic. Eventually, due to geodetic precession, the companion moved out of the line of sight, but a value for the mass ratio of $R = M_p/M_c = 1.0714 \pm 0.0011$ was still obtained[21].

Because of the fast spin, the timing precision is much better and the parameters improved in precision since. This allowed an entire six PK parameters to be measured for this system as shown in Fig. 4, all of which agree amazingly with GR! The following looks at four different predictions of GR and how they were tested with the double pulsar. Spoiler - GR passed all the tests perfectly!![12][18]

The main benefit is the closeness of this binary as compared to the HT binary, where the distance could even be measured using parallax, and it also experiences orbital decay like the HT pulsar. The intrinsic decay, $\dot{P_b^{int}}$, was found to be $-1.247752(79) \times 10^{-12}$ which agrees almost exactly with the value predicted by GR, the ratio being $\dot{P_b^{int}}/\dot{P_b^{GR}} = 0.999963(63)$ [18]. The masses were estimated using PK parameters. In Fig. 5, the decay in the orbital period is shown through the change in the periastron over time. This wouldn't be the case if the prediction of GWs by GR wasn't real, but it lines up perfectly with the model GR set up of how the decay would look. This is also an amazing test that shows the quadrupole formula for the emission of GW is valid.

The propagation of the signal is affected due to GR, as the relativistic Shapiro effect is strong in this edge-on binary, especially when the companion sits directly in the line of sight of the

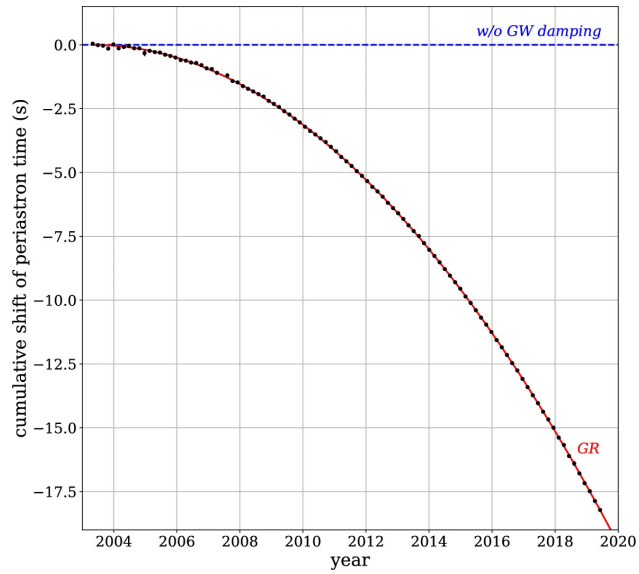

Figure 5: Periastron decay throughout time due to GWs, predicted by GR as compared to model without GR[18].

primary and the light must propagate through the curved spacetime, getting deflected by up to 0.03°. From timing the primary, the Shapiro parameters, r and s, could be known accurately, leading to the inclusion of PN corrections to the mass function to find the masses precisely. Previously, experiments like this were only carried out within the solar system, so this was a big step in testing GR in a strong field[22].

The precision in the periastron gives the strictest constraints, where the second order PN parameters are included as well as contributions from the Lense-Thirring effect. Additionally, as the system is edge-on, the primary gets occasionally eclipsed by the companion, so geodetic spin precession can be measured by studying the change in the eclipsing pattern[12]. More examples are given in the Appendix.

5 Conclusion

From the discovery of the first pulsar by Jocelyn Bell, to the discovery of the DNS binary by Hulse and Taylor, we have come a long way with the knowledge gleaned from the study of pulsars and binary systems. Various gravity theories have been proposed, but GR is particularly lovely in the predictions made being visible through the data collected. The quadrupole nature of gravity was predicted by GR, many binaries were able to put GR to the test, and it passed with flying colours[12]. Various relativistic effects, such as the Shapiro delay[14], Rømer delay[5], etc., were predicted and finally shown to exist in a strong-field, where they increased the accuracy of our ability to predict and the models used to describe such systems. The big breakthrough was the observed decay of the periastron of the double pulsar binary, which proved the existence of GWs, another excellent prediction by GR that

proved to be true[18]. The PK parameters assuming GR, once found, allow the uncovering of many other characteristics of the binary, ones that otherwise would remain unknown[18].

GR is pervasive in the world around us, and gravity, which otherwise has a very weak effect, could be studied in strong-field limits. The equivalence principle also holds, as with GR, the internal structure doesn't matter. With so many possibilities, general relativity remains the most probable and best description, beating most other theories in virtually all the tests discussed, as well as making incredibly accurate predictions.

Acknowledgements

I'd like to acknowledge Leyla Saraj, of whom I discussed with often to understand concepts such as gravitational waves, the various relativistic effects, and other intricacies that required a back-and-forth to comprehend. I'd also like to acknowledge our lovely TA, Divyesh Dagia, who was incredibly helpful during tutorial sessions with guiding us through problems when stuck, or offering hints to push us to think and figure it out ourselves. Of course, it goes without saying, a big thank you to our Professor, Dr. Czarnecki, for the wonderful class. There was a lot of content to learn and the class required dedication, but there was a lot of insight to be gained into understanding the universe around us.

6 Appendix

Abbreviations

CoM	Centre-of-Mass
DNS	Double Neutron Star
EoS	Equation of State
GR	General Relativity
GW	Gravitational Waves
HT	Hulse-Taylor
MoI	Moment of Inertia
MSP	Millisecond Pulsars
NS	Neutron Star
PK	Post-Keplerian
PN	Post-Newtonian
SNR	Signal-Noise Ratio
SSB	Solar System Barycentre
ToA	Time of Arrival
WD	White Dwarfs

Equations for PK parameters[13][23][15]

Shorthand: $s \equiv sin(i),\ X_p = M_p/M,\ X_c = M_c/M.$

$$\dot{\omega} = 3\left(\frac{P_b}{2\pi}\right)^{-5/3}(T_\odot M)^{2/3}(1-e^2)^{-1} \tag{10}$$

$$\gamma = \left(\frac{P_b}{2\pi}\right)^{1/3}(T_\odot M)^{2/3}X_c(1+X_c)e \tag{11}$$

$$\dot{P}_b = -\frac{192\pi}{5}\left(\frac{P_b}{2\pi}\right)^{-5/3}(T_\odot M)^{5/3}X_pX_c\frac{1+\frac{73}{24}e^2+\frac{37}{96}e^4}{(1-e^2)^{7/2}} \tag{12}$$

$$r = (T_\odot M)X_c = (T_\odot M_c) \tag{13}$$

$$s = x\left(\frac{P_b}{2\pi}\right)^{-2/3}(T_\odot M)^{-1/3}X_C^{-1} \tag{14}$$

$$\delta_\theta = \left(\frac{P_b}{2\pi}\right)^{-2/3}(T_\odot M)^{2/3}\left(\frac{7}{2}X_p^2+6X_pX_c+2X_c^2\right) \tag{15}$$

$$\delta_r = \left(\frac{P_b}{2\pi}\right)^{-2/3}(T_\odot M)^{2/3}\left(3X_p^2+6X_pX_c+2X_c^2\right) \tag{16}$$

Additional Binaries Testing GR

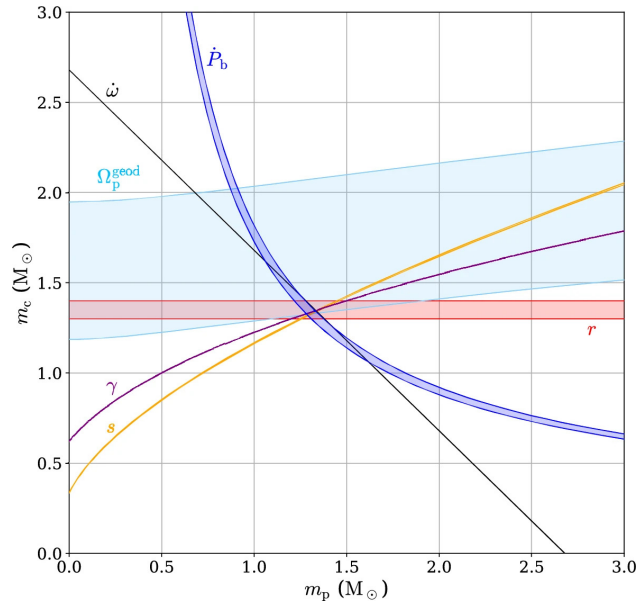

Figure 6: PSR J19061+0746: An edge-on MSP system with three PK parameters measured. Strong Shapiro delay allowed for two other parameters. This was the firs test ground for Shapiro delay in strong-field. Data from [24], [25].

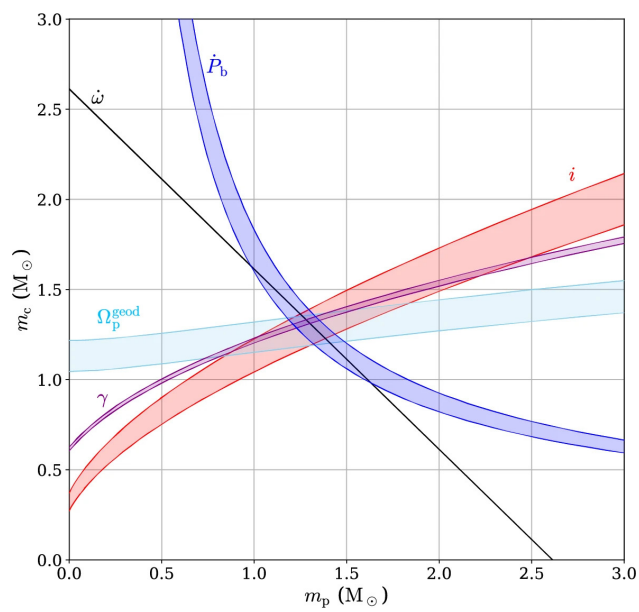

Figure 7: PSR J19061+0746: Non-recycled young pulsar, higher surface magnetic field and slower spin period. Three PK parameters detected, precession calculated was in perfect agreement with GR. Timing observations for PK parameters from [26], precession and inclination from [27].

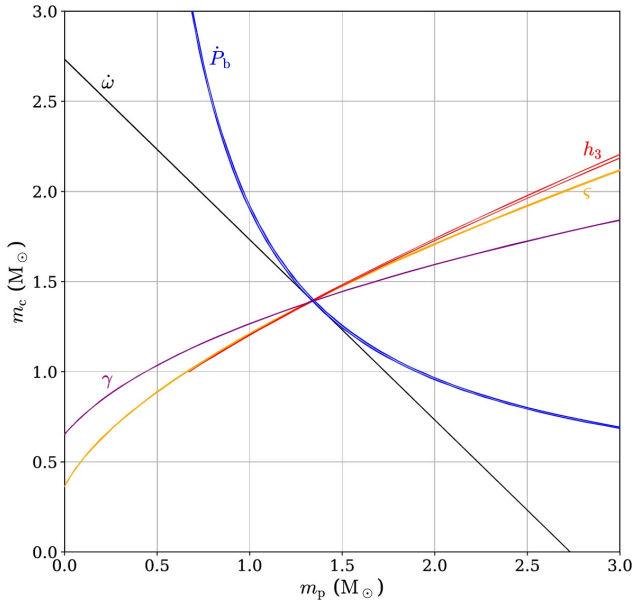

Figure 8: PSR J17571854: Similar to HT pulsar, but highly eccentric. Orbital period decayed much quicker. Five PK parameters measured but distance unknown, model was used. Parameter values from [28].

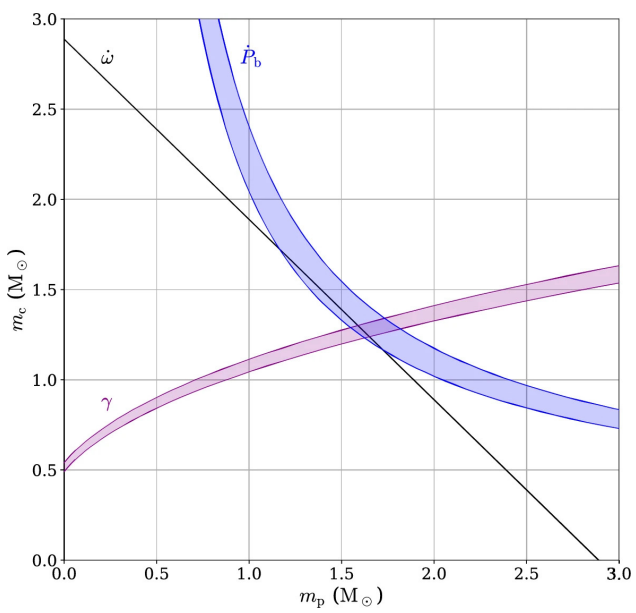

Figure 9: PSR J1913+1102: Three PK parameters of this binary measured, very asymmetric in mass, showing significant GW damping. Predicted GR value agrees with measured value. Figure from [29].

References

[1] R. A. Hulse and J. H. Taylor, "Discovery of a pulsar in a binary system," *Astrophysical Journal, vol. 195, Jan. 15, 1975, pt. 2, p. L51-L53.*, vol. 195, pp. L51–L53, 1975.

[2] B. P. Abbott et al., "Gw150914: The advanced ligo detectors in the era of first discoveries," *Physical review letters*, vol. 116, no. 13, p. 131 103, 2016.

[3] J. H. Taylor and J. M. Weisberg, "A new test of general relativity: Gravitational radiation and the binary pulsar psr 1913+16," *ApJ*, vol. 253, pp. 908–920, 1982. DOI: 10.1086/159690.

[4] A. Hewish, S. J. Bell, J. D. Pilkington, P. Frederick Scott, and R. A. Collins, "Observation of a rapidly pulsating radio source," in *A Source Book in Astronomy and Astrophysics, 1900–1975*, Harvard University Press, 1979, pp. 498–504.

[5] D. R. Lorimer and M. Kramer, *Handbook of Pulsar Astronomy*. Cambridge University Press, 2005.

[6] H. D. Craft Jr, *Radio observations of the pulse profiles and dispersion measures of twelve pulsars*. Cornell University, 1970.

[7] J. M. Cordes and T. J. W. Lazio, "NE2001: A new model for the galactic distribution of free electrons and its fluctuations," *arXiv e-prints*, 2002. eprint: astro-ph/0207156.

[8] J. M. Lattimer and M. Prakash, "Neutron star observations: Prognosis for equation of state constraints," *Phys. Rep.*, vol. 442, pp. 109–165, 2007. DOI: 10.1016/j.physrep.2007.02.003.

[9] R. T. Edwards, G. B. Hobbs, and R. N. Manchester, "Tempo2, a new pulsar timing package – i. an overview," *MNRAS*, vol. 372, pp. 1549–1574, 2006. DOI: 10.1111/j.1365-2966.2006.10903.x.

[10] D. Bhattacharya and E. P. J. van den Heuvel, "Formation and evolution of binary and millisecond radio pulsars," *Physics Reports*, vol. 203, pp. 1–124, 1991. DOI: 10.1016/0370-1573(91)90064-S.

[11] R. N. Manchester, G. B. Hobbs, A. Teoh, and M. Hobbs, "The australia telescope national facility pulsar catalogue," *The Astronomical Journal*, vol. 129, no. 4, p. 1993, 2005.

[12] P. C. Freire and N. Wex, "Gravity experiments with radio pulsars," *Living Reviews in Relativity*, vol. 27, no. 1, p. 5, 2024.

[13] T. Damour and N. Deruelle, "General relativistic celestial mechanics of binary systems. ii. the post-newtonian timing formula," in *Annales de l'IHP Physique théorique*, vol. 44, 1986, pp. 263–292.

[14] I. I. Shapiro, "Fourth test of general relativity," *Phys. Rev. Lett.*, vol. 13, pp. 789–791, 1964. DOI: 10.1103/PhysRevLett.13.789.

[15] T. Damour and J. H. Taylor, "Strong-field tests of relativistic gravity and binary pulsars," *Physical Review D*, vol. 45, no. 6, p. 1840, 1992.

[16] P. C. Peters and J. Mathews, "Gravitational radiation from point masses in a keplerian orbit," *Phys. Rev.*, vol. 131, pp. 435–440, 1963. DOI: 10.1103/PhysRev.131.435.

[17] P. C. C. Freire and N. Wex, "Tests of gravitational theories with pulsar binary systems," *Living Reviews in Relativity*, vol. 13, p. 7, 2010. DOI: 10.12942/lrr-2010-7.

[18] M. Kramer et al., "Strong-field gravity tests with the double pulsar," *Physical Review X*, vol. 11, no. 4, p. 041 050, 2021.

[19] J. M. Weisberg and Y. Huang, "Relativistic measurements from timing the binary pulsar psr b1913+ 16," *The Astrophysical Journal*, vol. 829, no. 1, p. 55, 2016.

[20] J. M. Weisberg and Y. Huang, "Relativistic measurements from timing the binary pulsar psr b1913+16," *ApJ*, vol. 829, p. 55, 2016. DOI: 10.3847/0004-637X/829/1/55.

[21] M. Burgay et al., "An increased estimate of the merger rate of double neutron stars from observations of a highly relativistic system," *Nature*, vol. 426, pp. 531–533, 2003. DOI: 10.1038/nature02124.

[22] M. Kramer et al., "Tests of general relativity from timing the double pulsar," *science*, vol. 314, no. 5796, pp. 97–102, 2006.

[23] J. H. Taylor and J. M. Weisberg, "Further experimental tests of relativistic gravity using the binary pulsar psr 1913+ 16," *Astrophysical Journal, Part 1 (ISSN 0004-637X), vol. 345, Oct. 1, 1989, p. 434-450. Research supported by NSF.*, vol. 345, pp. 434–450, 1989.

[24] E. Fonseca, I. H. Stairs, and S. E. Thorsett, "A comprehensive study of relativistic gravity using psr b1534+ 12," *The Astrophysical Journal*, vol. 787, no. 1, p. 82, 2014.

[25] H. Ding, A. T. Deller, E. Fonseca, I. H. Stairs, B. Stappers, and A. Lyne, "The orbital-decay test of general relativity to the 2% level with 6 yr vlba astrometry of the double neutron star psr j1537+ 1155," *The Astrophysical Journal Letters*, vol. 921, no. 1, p. L19, 2021.

[26] B. Stappers et al., "The binary companion of young, relativistic pulsar j1906+ 0746," *apj*, vol. 798, 2015.

[27] G. Desvignes et al., "Radio emission from a pulsar's magnetic pole revealed by general relativity," *Science*, vol. 365, no. 6457, pp. 1013–1017, 2019.

[28] A. Cameron et al., "New constraints on the kinematic, relativistic, and evolutionary properties of the psr j1757- 1854 double neutron star system," *Monthly Notices of the Royal Astronomical Society*, vol. 523, no. 4, pp. 5064–5085, 2023.

[29] R. Ferdman et al., "Asymmetric mass ratios for bright double neutron-star mergers," *Nature*, vol. 583, no. 7815, pp. 211–214, 2020.